

~~***~~ Important Python string datatype concept

6 June

* A single or group of characters which are enclosed in between quotations are called as string in python

* A string is an ordered collection of characters which can include letters, numbers, symbols and spaces

--> For Example `s1 = "H"` `s2 = "Hello"` `s3 = 'welcome'`

-> by using single or double quotations we can create single line string

```
s1 = 'srinivas'
```

```
s2 = "Balaji"
```

→ by using triple single or triple double quotations we can create multiline string

```
s1 = '''Line1
```

```
Line2
```

```
Line3'''
```

```
print(s)
```

* string datatype represents using str type

```
s = 'srinivas'
```

```
print(type(s))
```

~~***~~ we can create string object by using empty quotations or str() function for example

```
s1 = "" s2 = str()
```

- * String is Imutable type object (not editable object)
- * String objects follows insertion order
- * String allows duplicate elements
- * Every character in the string object is represented with unique index
- * Python supports both forward and backward index
- * Forward indexing starts from 0 and it representing the first element of the given object
- * Backward indexing starts from -1 and 99 representing the last element of the given object

→ For Example

0	1	2	3	4	5	6	7
S	✓	I	n	I	✓	A	S
0	1	2	3	4	5	6	7

$S[0] = S$ $S[1] = \checkmark$ $S[7] = S$ $(S[8])$
|
Index error

→ Slicing → object [start - index - value : end - index - value : step - value]

* By using slicing syntax is important

Here,

* by default start index value is 0

* by default end index value is given object length

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→ For Example

0	1	2	3	4	5	6	7
S	V	I	n	I	V	A	S
0	1	2	3	4	5	6	7

$S[0] = S$ $S[1] = V$ $S[7] = S$ $S[8]$

Index error

→ Slicing → object [start - index - value : end - index - value : step - value]

* By using slicing syntax is important

Here,

* by default start index value is 0

* by default end index value is given object length

By default step value is 1
\$[0:8:1]\$

(str)

Python string supports "Concatenation" behaviour

'sri' + 'vas'

'sriivas'

Python string supports "repetition" behaviour

"Python" * 3

Output: "PythonPythonPython"

Python string supports string unpacking

a, b, c = 'sri'

a
"s"

b
"r"

c
"i"

Python string supports membership operator

Python string methods

Function name (object)

object . method name

~~object~~

1) upper

object . upper ()

2) Lower

object . lower ()

June 9

⑤ capitalize

object.capitalize()

⑥ title

object.title()

* swapcase

object.swapcase()

⑦

* object.split() --> default considered as space

object.split("delimiter")

object.splitlines

object.replace()

object.join()

object.startswith (substring)

object.endswith (substring)

object.strip()

object.rstrip()

object.isalnum()

object.isupper()

object.islower()